ARM64 vs x86-64 HPC Research Topics

MSc Thesis Ideas in High Performance Computing and Computer Architecture

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1. Performance-per-Watt in HPC

Research Question

How does ARM64 compare to x86-64 in performance-per-watt for scientific HPC workloads?

Abstract

This research evaluates whether modern ARM64 processors can provide superior energy efficiency compared to x86-64 systems for scientific HPC applications. The study benchmarks numerical and parallel workloads to analyze tradeoffs between execution performance, power consumption, and scalability.

Tooling

LINPACK, HPCG, Linux perf, RAPL, ARM Streamline

References

- Simakov et al., Are We Ready for Broader Adoption of ARM in the HPC Community?, ACM HPC Asia Workshops, 2023.
- Elwasif et al., Application Experiences on a GPU-Accelerated Arm-Based HPC Testbed, ACM HPC Asia Workshops, 2023.

2. ARM CPU-GPU HPC Systems

Research Question

Can ARM-based CPU-GPU systems provide competitive performance for heterogeneous HPC applications?

Abstract

This thesis investigates the performance characteristics of ARM-based CPU-GPU systems for heterogeneous HPC workloads. The research focuses on CPU-GPU coordination overheads, memory behavior, and application scalability compared to traditional x86-64-based platforms.

Tooling

CUDA, NVIDIA Nsight Systems, Slurm

References

- Elwasif et al., Application Experiences on a GPU-Accelerated Arm-Based HPC Testbed, ACM HPC Asia Workshops, 2023.
- Simakov et al., First Impressions of the NVIDIA Grace CPU Superchip and NVIDIA Grace Hopper Superchip, ACM HPC Asia Workshops, 2024.

3. ARM SVE vs AVX-512

Research Question

How does vectorization performance differ between ARM SVE and x86 AVX-512 architectures?

Abstract

This research compares the effectiveness of ARM Scalable Vector Extension (SVE) and x86 AVX-512 vector instruction sets for scientific computing workloads. The study evaluates vectorization efficiency, compiler optimization behavior, and application throughput across architectures.

Tooling

LLVM/Clang, GCC, ARM SVE toolchains, Intel VTune

References

- Evaluation of Vectorization Methods on Arm SVE Using the Exo Language, IEEE Cluster Workshops, 2024.
- Simakov et al., Are We Ready for Broader Adoption of ARM in the HPC Community?, ACM HPC Asia Workshops, 2023.

4. Memory-Centric Profiling

Research Question

Can memory-centric profiling improve optimization of ARM-based HPC systems?

Abstract

This thesis investigates whether memory-centric profiling techniques can identify bottlenecks and improve optimization strategies for ARM-based HPC systems. The research focuses on cache utilization, memory locality, and bandwidth behavior in scientific applications.

Tooling

perf, ARM SPE, PAPI, STREAM benchmark

References

- Multi-level Memory-Centric Profiling on ARM Processors with ARM SPE, IEEE/ACM SC Workshops, 2024.
- Simakov et al., First Impressions of the NVIDIA Grace CPU Superchip and NVIDIA Grace Hopper Superchip, ACM HPC Asia Workshops, 2024.

5. Transformer Inference on ARM

Research Question

How effective are ARM many-core CPUs for transformer inference workloads?

Abstract

This research evaluates the suitability of ARM many-core processors for transformer inference and AI-HPC hybrid applications. The study examines optimization strategies, scalability, and throughput compared to traditional x86-64 systems.

Tooling

PyTorch, ONNX Runtime, Linux perf, Hugging Face Transformers

References

- Jiang et al., Full-Stack Optimizing Transformer Inference on ARM Many-Core CPU, IEEE TPDS, 2023.
- Simakov et al., First Impressions of the NVIDIA Grace CPU Superchip and NVIDIA Grace Hopper Superchip, ACM HPC Asia Workshops, 2024.

6. AI-Assisted Architecture Tuning

Research Question

Can AI-assisted architecture tuning improve performance optimization on ARM HPC systems?

Abstract

This thesis investigates whether AI-assisted optimization techniques can improve architecture tuning and workload configuration for ARM-based HPC systems. The research evaluates automated tuning methods for performance and energy efficiency improvements.

Tooling

Python ML libraries, Bayesian optimization frameworks, autotuning tools

References

- AI-Assisted Design-Space Analysis of High-Performance Arm, IEEE/ACM SC Workshops, 2024.
- Simakov et al., Are We Ready for Broader Adoption of ARM in the HPC Community?, ACM HPC Asia Workshops, 2023.

7. Memory-Bound Parallel Applications

Research Question

How do ARM64 and x86-64 architectures compare for memory-bound parallel applications?

Abstract

This study compares ARM64 and x86-64 systems under memory-intensive parallel workloads commonly found in scientific computing. The research focuses on memory bandwidth utilization, cache behavior, and parallel scalability.

Tooling

STREAM benchmark, OpenMP, HPCToolkit

References

- Multi-level Memory-Centric Profiling on ARM Processors with ARM SPE, IEEE/ACM SC Workshops, 2024.
- Elwasif et al., Application Experiences on a GPU-Accelerated Arm-Based HPC Testbed, ACM HPC Asia Workshops, 2023.

8. Energy-Efficient ARM Datacenters

Research Question

Can ARM-based HPC systems reduce datacenter energy consumption without significant performance loss?

Abstract

This thesis evaluates whether ARM-based HPC systems can improve energy efficiency at the datacenter level while maintaining acceptable computational performance. The study benchmarks scientific workloads under sustained utilization scenarios.

Tooling

Prometheus, Grafana, RAPL, Slurm accounting

References

- Simakov et al., Are We Ready for Broader Adoption of ARM in the HPC Community?, ACM HPC Asia Workshops, 2023.
- Simakov et al., First Impressions of the NVIDIA Grace CPU Superchip and NVIDIA Grace Hopper Superchip, ACM HPC Asia Workshops, 2024.

9. Distributed HPC on ARM

Research Question

How effective are ARM-based systems for GPU-accelerated distributed HPC applications?

Abstract

This research investigates the scalability and communication efficiency of distributed HPC workloads running on ARM-based GPU-accelerated systems. The study evaluates MPI communication overheads and application scaling characteristics.

Tooling

Open MPI, NCCL, Score-P, OSU benchmarks

References

- Elwasif et al., Application Experiences on a GPU-Accelerated Arm-Based HPC Testbed, ACM HPC Asia Workshops, 2023.
- Simakov et al., First Impressions of the NVIDIA Grace CPU Superchip and NVIDIA Grace Hopper Superchip, ACM HPC Asia Workshops, 2024.

10. Portability of HPC Optimizations

Research Question

How portable are HPC optimization techniques between ARM64 and x86-64 architectures?

Abstract

This thesis studies whether HPC optimization strategies developed for x86-64 systems can be effectively transferred to ARM64 architectures. The research analyzes portability challenges involving vectorization, memory optimization, and compiler tuning.

Tooling

LLVM/Clang, GCC, OpenMP, SPEC CPU

References

- Evaluation of Vectorization Methods on Arm SVE Using the Exo Language, IEEE Cluster Workshops, 2024.
- Jiang et al., Full-Stack Optimizing Transformer Inference on ARM Many-Core CPU, IEEE TPDS, 2023.

Any Questions?